Multiple Discrete Random Variables

Lecture 5

Topics in this lecture

- Joint PMF
- Conditional PMF
- Conditional expectation
- Independence of random variables

Joint PMF

- Let X and Y be discrete random variables, the joint PMF of X and Y, denoted by $p_{X,Y}(x,y)$, captures the probabilities of the events in the form of $\{X = x,\ Y = y\}$

- Joint probability mass function

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

Compute Probability from Joint PMF

- Finding probability of event A

$$P(A) = \sum_{(x,y)\in A} p_{X,Y}(x, y)$$

- Sum over all values of joint pmf

$$\sum_{\forall (x,y)} p_{X,Y}(x, y) = 1$$

Example 5.1

- Let X and Y be the marks that show up after rolling two 4-faced dice. The joint PMF $p_{X,Y}(x,y)$ is given in the following table. Find the probability of an event $\{X > 3, Y < 3\}$

Y \ X	1	2	3	4
1	1/24	2/24	2/24	1/24
2	1/24	2/24	2/24	1/24
3	1/24	1/24	1/24	1/24
4	2/24	2/24	2/24	2/24

Example

10คน

- A statistic shows that 10% of people entering a store will buy an iPhone, 30% will buy an iPad, and 60% will not buy anything. If 10 people enter the store today, what is the probability that exactly 2 iPhone and 2 iPad will be sold?

30คน 60คน

เข้า 100คน

ซื้อ phone 10 คน

pad 30 คน

ไม่ซื้อ 60 คน

0.02 0.04

0.03

Marginal PMF

$$p_X(x) = P(X = x)$$

$$= P(X = x, Y = \text{anything})$$

$$= \sum_y p_{X,Y}(x, y)$$

Similarly

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Example 5.2

- Let X and Y be the marks that show up after rolling two 4-faced dice. The joint PMF $p_{X,Y}(x,y)$ is given in the following table. Find the marginal PMF of X.

Y \ X	1	2	3	4
1	1/24	2/24	2/24	1/24
2	1/24	2/24	2/24	1/24
3	1/24	1/24	1/24	1/24
4	2/24	2/24	2/24	2/24

Conditional PMF

- Conditional PMF on Event A

$$p_X(x|A) = P(X = x|A) = \frac{P(\{X = x\} \cap A)}{P(A)} \quad , \quad P(A) > 0$$

- Conditional PMF

$$p_{X|Y}(x|y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

Example 5.3

- Let X and Y be the marks that show up after rolling two 4-faced dice. The joint PMF $p_{X,Y}(x,y)$ is given in the following table. Find the conditional PMF of $X|Y=2$.

X (columns), Y (rows)

	1	2	3	4
1	1/24	2/24	2/24	1/24
2	1/24	2/24	2/24	1/24
3	1/24	1/24	1/24	1/24
4	2/24	2/24	2/24	2/24

Example

- Pick a number X at random from a set {1, 2, 3, 4, 5}. Then, pick another random number Y, where Y is a positive integer that is less than or equal to X.

    - Find the joint PMF of X, Y

    - Find the conditional PMF of $X \mid Y = 3$

Conditional Expectation

- Conditional expectation of X|Y = y

$$E[X|Y = y] = \sum_{x} x \, p_{X|Y}(x|y)$$

- Finding E[X] from conditional expectation

$$E[X] = E[E[X|Y]] = \sum_{y} E[X|Y = y] \, p_Y(y)$$

Example 5.4

- Let X and Y be the marks that show up after rolling two 4-faced dice. The joint PMF $p_{X,Y}(x,y)$ is given in the following table. Find the conditional expectation $E[X|Y=2]$.

X (columns), Y (rows)

	1	2	3	4
1	1/24	2/24	2/24	1/24
2	1/24	2/24	2/24	1/24
3	1/24	1/24	1/24	1/24
4	2/24	2/24	2/24	2/24

Example

- Pick a number X at random from a set {1, 2, 3, 4, 5}. Then, pick another random number Y, where Y is a positive integer that is less than or equal to X.
    - Find the conditional expectation $E[X \mid Y = 3]$

Independence of 2 RV

- If *X* and *Y* are independent discrete random variables, then the joint pmf is equal to the product of the marginal pmf

$$p_{X,Y}(x,y) = p_X(x)p_Y(y) \quad , \quad \forall x, y$$

Example 5.5

- Let X and Y be the marks that show up after rolling two 4-faced dice. The joint PMF $p_{X,Y}(x,y)$ is given in the following table. Are X, Y independent?

X

Y	1	2	3	4
1	1/24	2/24	2/24	1/24
2	1/24	2/24	2/24	1/24
3	1/24	1/24	1/24	1/24
4	2/24	2/24	2/24	2/24